

Department of Leather Engineering
Khulna University of Engineering & Technology
B.Sc. Engineering 3rd Year 2nd Term Examination, 2019
LE 3209

Leather Products Engineering II

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION-A

- | | | |
|------|---|----|
| 1(a) | Write down the objectives and required quality of lining materials for leather products. | 08 |
| 1(b) | Distinguish among interlining, interfacing and underlining. | 10 |
| 1(c) | Briefly-Explain the origin, properties and uses of "SILK" as lining materials. | 10 |
| 1(d) | Write down the short notes on mercerization. | 07 |
| | | |
| 2(a) | Differentiate between woven and non-woven fabric for leather products. | 06 |
| 2(b) | What is fabrication technology? Mention the major stages of fabrication technology with example. | 07 |
| 2(c) | Why creasing is done? Write down the instructions and acceptance criteria of creasing for leather products. | 10 |
| 2(d) | What is cementing? State the process control, instruction and acceptance criteria of folding. | 12 |
| | | |
| 3(a) | What are the differences between chain stitch and lock stitch? | 06 |
| 3(b) | Define thread. Briefly discuss the thread sizing systems. | 13 |
| 3(c) | Write down the short notes on:
i. Closed seam ii. Butted seam iii. Cable stitching and iv. Seam puckering | 12 |
| 3(d) | Which factors are significant that influence on the thread demand? | 04 |
| | | |
| 4(a) | Why cutting point is suitable for sewing needle in leather products stitching? | 05 |
| 4(b) | List out the major joining techniques used in leather products manufacturing. | 04 |
| 4(c) | Write down the goals of material handling for leather products manufacturing. | 06 |
| 4(d) | Differentiate between assembly line system and conveyor system. Explain which one is more suitable for leather goods manufacturing. | 12 |
| 4(e) | Write the short notes on i. Felt and ii. Rayon lining | 08 |

SECTION-B

5(a)	What is spec-sheet? Make a blueprint of a specification sheet for “Fashionable Ladies Bag”.	11
5(b)	Briefly explain the major steps in new product development.	16
5(c)	Define the concept board. List out the predicting steps are considered during making a concept board.	08
6(a)	What are the acceptance criteria for fancy leather goods manufacturing?	05
6(b)	Explain the assembling process of a key case for the following steps: i. Hook piece preparation ii. Asther preparation iii. Final preparation	15
6(c)	Define reinforcement. Write the influencing factors of fusing materials used in leather products industry.	11
6(d)	List out the consumable materials in bag manufacturing.	04
7(a)	Define feed. Explain the different types of feeding systems are used in a conventional sewing machine.	12
7(b)	What is sewing? Describe the sewing faults and their possible remedies.	13
7(c)	Mention the functions of the following parts of the sewing machine: i. Knee lift ii. Check spring and iii. Needle bar	06
7(d)	Why edge construction is important according to distinguish leather product manufacturing?	04
8(a)	Define Bag. State the features and functions for the following bags with related figure: i. Bag pack ii. Tote bag iii. Barrel bag and iv. Satchel bag	14
8(b)	What is gusset? Explain the three part “V” gusset preparation.	06
8(c)	Write short notes on i. Zipper and ii. Edge construction	08
8(d)	Cite the two significant methods of fusing in leather garments manufacturing.	07

Department of Leather Engineering
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 B. Sc. Engineering 3rd Year 2nd Term Examination-2019
Leather Microbiology
LE 3219

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
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 iii) Assume reasonable data if any missing.

SECTION-A

- | | | |
|------|---|----|
| 1(a) | Write down the chemical structure of nucleoid and ribosome of a bacterial cell. | 08 |
| 1(b) | Discuss the growth requirements factors for bacteria. | 08 |
| 1(c) | Prove with a figure, $n = \frac{\text{Logb} - \text{Loga}}{\text{Log2}}$ where n is the number of bacterial generation. | 12 |
| 1(d) | Mention the role of bacteria in the tanning industry. | 07 |
| 2(a) | Write down the sporulation process of bacteria with a necessary figure. | 08 |
| 2(b) | If bacterial spores are able to resist harsh conditions, how do we destroy them? | 10 |
| 2(c) | Mention five chemical agents with their required concentrations and times for chemical destruction of selected microbes. | 10 |
| 2(d) | What are infections and pathogens? How can infectious disease transmit? | 07 |
| 3(a) | What is meant by microbial colony? Write down the chemical composition and application of the following culture media: i. Differential media ii. Synthetic media | 12 |
| 3(b) | Suppose you have been given a mixed colony agar plate cultured from the source of liming wastewater. Now you are said to isolate a specific colony from it. Write down the sequential steps of the isolation process. | 10 |
| 3(c) | List out four microbial diseases and their resulting defects in leather. | 08 |
| 3(d) | Distinguish between gram-positive and gram-negative bacteria. | 05 |
| 4(a) | State the principle of gram staining method. Why microbacterium can not be stained by using this method? | 11 |
| 4(b) | What are biocides? Write down the name of three biocides with their chemical structure and applications in leather processing. | 10 |
| 4(c) | In spite of being pathogenic, why do we need bacteria? | 06 |
| 4(d) | What is anthrax? Briefly discuss the modes of infection and transmission of anthrax. | 08 |

SECTION-B

5(a)	Differentiate between the morphological features of yeasts and moulds.	08
5(b)	Briefly discuss the sexual reproduction process in fungi.	10
5(c)	Write notes on any two non-selective culture media for the laboratory cultivation of fungus.	08
5(d)	Write about the possible microbiological problems in tanning industry during storage of hides/skins with the remedies.	09
6(a)	With the help of the Lineweaver-Burk plot state the nature of kinetics parameters for different enzyme inhibitors.	09
6(b)	Elucidate the transcription process of enzyme biosynthesis in prokaryotes.	12
6(c)	Show the mechanism of combined allosteric and covalent modification method of enzyme activity regulation.	06
6(d)	What is screening in microbiology? What sorts of work should be done in secondary screening?	08
7(a)	State the concept of fermentation. What are the ranges of industrial fermentation?	08
7(b)	State the role of the microbial growth phase in the fermentation process.	08
7(c)	List out the unit operations involved in upstream and downstream processing of industrial fermentation.	06
7(d)	What is fed-batch fermentation? Give the advantages and limitations of the process.	07
7(e)	Write down the ideal features of a bioreactor.	06
8(a)	Define the following terms: i. Enzyme activity ii. Specific activity and iii. Turnover number	07
8(b)	Write down the mechanism of physical adsorption and crosslinking method of enzyme immobilization.	10
8(c)	Explain shortly the nutrition tasting process of fungi.	10
8(d)	Write down the fermentation pathway of alcohol production from glucose.	08

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 2nd Term Examination-2019
Manufacturing Process
ME 3221

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
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 iii) Assume reasonable data if any missing.

SECTION-A

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|--|----|
| 1(a) Define casting. Describe basic requirements of casting. | 09 |
| 1(b) Briefly describe various patterns on the basis of physical structure. | 13 |
| 1(c) Explain steel moulding process with neat sketches. | 13 |
| 2(a) Define and classify die casting. Also illustrate hot chamber die casting method with neat sketch. | 15 |
| 2(b) Describe the "lost-wax" casting method with necessary sketches. Write down the application of "lost-wax" method. | 15 |
| 2(c) Write down the applications of casting. | 05 |
| 3(a) Define welding. Describe electric arc welding and gas welding in details. | 15 |
| 3(b) What is meant by metal forming process? Write down the differences between hot working and cold working process. | 12 |
| 3(c) Is it possible to do hot working process in room temperature for same metals? How. List the metals suitable for cold working process. | 08 |
| 4(a) Explain direct extrusion method. Also state its advantages and limitations. | 10 |
| 4(b) Differentiate between blanking & punching operation and coining & embossing operation. | 11 |
| 4(c) Describe the importance of metal forming operation in leather and footwear manufacturing. | 14 |

SECTION-B

- | | |
|--|----|
| 5(a) Describe geometry of a single point cutting tool with figure. | 12 |
| 5(b) Describe the mechanism of chip formation. Explain different types of chip with necessary sketches. | 15 |
| 5(c) Distinguish between orthogonal and oblique cutting. | 08 |
| 6(a) Draw the schematic diagram of a lathe machine with appropriate labeling. | 15 |
| 6(b) Explain the operation which can be performed on lathe machine. | 10 |
| 6(c) Write down the differences between a planer and a shaper. | 10 |
| 7(a) Describe quick return mechanism of a shaper machine. | 10 |
| 7(b) Differentiate between drilling and boring. | 06 |
| 7(c) Write short note on i. Counter boring ii. Counter sinking and iii. Spot facing. | 09 |
| 7(d) Describe simple indexing method with necessary sketches. | 10 |
| 8(a) Differentiate between conventional milling and climb milling. | 06 |
| 8(b) Describe centerless grinding with appropriate sketches. Also mention the advantages of centerless grinding. | 15 |
| 8(c) Write short note on i. Honing ii. Lapping and iii. Super finishing. | 09 |
| 8(d) Explain how this course materials are related to Leather Engineering. | 05 |

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 2nd Term Examination-2019
Footwear Engineering II
LE 3211

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
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 iii) Assume reasonable data if any missing.

SECTION-A

- | | |
|--|----|
| 1(a) Define three piece shoe last. | 03 |
| 1(b) State the traditional shoe last designing process by copying toe shape from an existing shoe. | 15 |
| 1(c) Classify shoe last based on material used. | 09 |
| 1(d) Mention the name of shoe constructions for different types of last bottom plates. | 08 |
| 2(a) Describe in brief the stretch joint last with figure. | 07 |
| 2(b) Define the following shoe last terminologies: i. Stick length ii. Crown and iii. Waist girth | 12 |
| 2(c) State the features of Oxford shoe last and moccasin shoe last. | 08 |
| 2(d) Make a comparison between straight grading and coordination grading. | 08 |
| 3(a) Define cutting value of leather. | 04 |
| 3(b) State the basic principles of synthetic cutting in a footwear industry. | 05 |
| 3(c) Describe briefly about travelling head clicking machine including its advantages and disadvantages. | 10 |
| 3(d) Make a sample job sheet for cutting department of a footwear industry. | 08 |
| 3(e) Write short notes on: i. Strap cutting machine and ii. Beam press machine | 08 |
| 4(a) What is meant by identification marking? Briefly discuss about the individual marking process. | 07 |
| 4(b) Mention the name of stitch marking methods. Illustrate the ways to stitch mark by press knife marking method. | 10 |
| 4(c) Which points are needed to watch during shoe upper punching operation? | 06 |
| 4(d) Sketch and describe the following seams: i. Bordo seam ii. Blind seam and iii. Sprung seam | 12 |

SECTION-B

- | | |
|---|----|
| 5(a) Define lasting. State the considering factors of lasting. | 09 |
| 5(b) What is shape retention? Explain the importance of plasticity and elasticity property of upper materials for shoe manufacturing. | 11 |
| 5(c) Classify the basic methods of lasting. Draw the cross sectional view of:
i. rib-welted ii. Californian and iii. McKay stitching methods of lasting. | 15 |
| 6(a) Write down the flow chart for Veldtschoen shoe construction. | 10 |
| 6(b) Concisely illustrate the back part moulding process of shoe. | 08 |
| 6(c) Insole attachment varies among different shoe constructions-Justify it. | 10 |
| 6(d) What is the special feature of contact mulling? Write down the advantages and disadvantages of force lasted shoe construction. | 07 |
| 7(a) Describe the forepart lasting process of shoe for cemented construction. | 09 |
| 7(b) Write down the important quality check points of lasted upper. | 08 |
| 7(c) Why heat setting is important? Describe the available strategies of heat setting operations. | 12 |
| 7(d) Welt is not a compulsory part in all shoe construction methods-explain. | 06 |
| 8(a) Preparation of bottom surface of lasted upper varies with the variation of upper materials and shoe construction-Explain. | 09 |
| 8(b) Differentiate among direct vulcanization, PVC injection and reaction PU moulding techniques of shoe construction. | 09 |
| 8(c) Briefly describe the methods of string attachment along the lasting edge of shoe upper for string lasting process. | 08 |
| 8(d) Explain following operations of Goodyear welted shoe construction:
i. Piece sole-fixing ii. Inseam trimming and iii. Stitch separating. | 09 |

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ME 3219

Fluid Mechanics and Machineries

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
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SECTION-A

- | | | |
|------|---|----|
| 1(a) | Define: Bulk Modulus of elasticity and Thixotropic fluid. | 06 |
| 1(b) | Fluid is considered as continuum. Justify it. | 06 |
| 1(c) | How pressure at a point varies with the orientation of the plane passing through a point? | 13 |
| 1(d) | The velocity distribution for the flow of a Newtonian fluid between two wide, parallel plates (Fig. | 10 |

01) is given by $u = \frac{3V}{2} \left[1 - \left(\frac{y}{h} \right)^2 \right]$

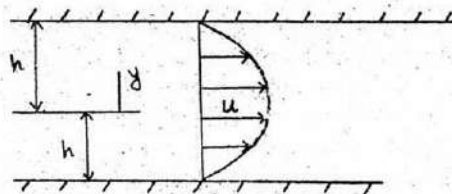


Fig. 01

Where V is the mean velocity. The fluid has a viscosity of 0.04 lb.s/ft². When V = 2 ft./s and h = 0.2 in. Determine:

- i. The shearing stress acting on the bottom wall and
- ii. The shearing stress acting on a plane to the walls and passing through the centerline (midplane).

- | | | |
|------|---|----|
| 2(a) | What are the various devices adopted for measuring fluid pressure? | 06 |
| 2(b) | Show that pressure is same in all directions at a point in a static fluid. | 07 |
| 2(c) | Explain how vacuum pressure can be measured with the help of a U-tube manometer. | 10 |
| 2(d) | The manometer fluid in the manometer of the following Fig. 02 has a specific gravity of 3.46. Pipe A and B both contain water. If the pressure in pipe A is decreased by 187.2 pa and pressure in pipe B increase by 129.6 Pa, determine the new differential reading of the manometer. | 12 |

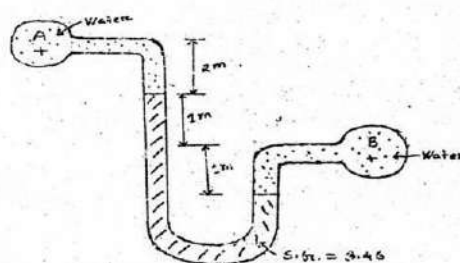


Fig. 02

- | | | |
|------|---|----|
| 3(a) | Draw a pressure diagram on a horizontal and vertical plane surface which is fully submerged into a stationary fluid. | 08 |
| 3(b) | Show that the centre of pressure of any lamina immersed under liquid is always below its centroid. | 12 |
| 3(c) | An open tank has a vertical partition and on one side contains gasoline with a density $\rho = 700 \text{ kg/m}^3$ at a depth of 4m, as shown in Fig. 03. A rectangular gate that is 4m high and 2m wide and hinged at one end is located in the partition. Water is slowly added to the empty side of the tank. At what depth, h, will the gate start to open? | 15 |

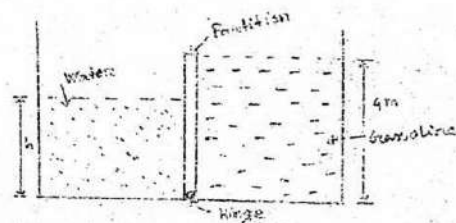
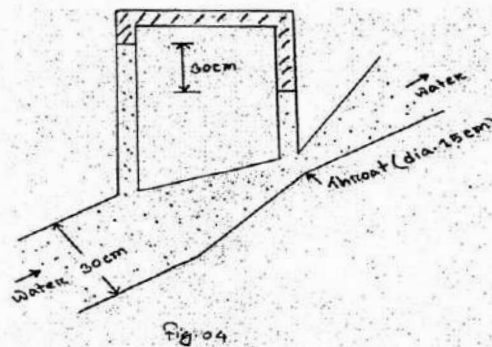


Fig. 03

- 4(a) Define metacentric height. Explain the stability condition in case of the partially submerged body with a schematic diagram. 15
- 4(b) Determine buoyant forces on a body floating at the surface of separation between two fluids. 07
- 4(c) A cylindrical buoy is 2m in diameter and 2.5 m long and weighs 2.2 metric tons. The density of seawater is 1025 kg/m^3 . Show that the buoy does not float with its axis vertical: What minimum pull should be applied to a chain attached to the center to keep it vertical? 13

SECTION-B

- 5(a) Differentiate between Lagrangian and Eulerian methods in fluid mechanics. 04
- 5(b) Discuss the characteristics of laminar flow, turbulent flow and rotational flow in brief. 06
- 5(c) What is venturimeter? Derive an expression for the discharge through a venturimeter. 12
- 5(d) Find the discharge of water flowing through a pipe 30 cm diameter placed in an inclined position where a venturimeter is inserted, having a throat-diameter of 15 cm. The pressure difference between the main and throat measured by an inverted U-tube manometer in Fig.04. The loss of head between main and the throat is 0.2 times the Kinetic head of the pipe. 13



- 6(a) Derive Euler's equation of motion with necessary assumptions. 12
- 6(b) Distinguish between fan, blower and compressor. 08
- 6(c) Discuss the classification of compressor. Also, write the working principle of the screw compressor. 15
- 7(a) Distinguish between single acting and double acting reciprocating pump with neat sketch. 10
- 7(b) Explain the functions of air vessels in a reciprocating pump. Show that by fitting air vessel 84.8% work can be saved against friction. 15
- 7(c) A double acting reciprocating pump having piston area 0.1 m^2 has a stroke 0.3 m long. The pump discharge 2.4 m^3 of water per minute at 45 rpm through a height 10 m. Find the slip of the pump and the power required to drive the pump. 10
- 8(a) Explain the working principle of a centrifugal pump with a neat sketch. 13
- 8(b) Discuss the conditions under which cavitation will not occur. 10
- 8(c) Derive the expression for work done by the impeller of a centrifugal pump. 12